

Adı Soyadı: CEVAP

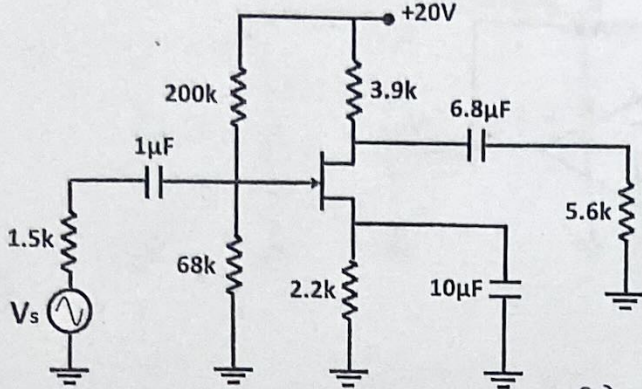
Numarası: ANAHTARI

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1 2 3 4

Tarih: 10.06.2022 Süre: 90 dak. BAŞARILAR... Prof.Dr.Seral ÖZŞEN & Doç.Dr. Murat CEYLAN

S.1.



$$I_{DSS}=10\text{mA}, V_p=-6\text{V}, g_m=1.95\text{mS}$$

$$C_{w1}=4\text{pF}, C_{w2}=6\text{pF}$$

$$C_{ds}=3\text{pF}, C_{gs}=12\text{pF}, C_{gd}=8\text{pF} \text{ ise}$$

a) $A_v = ?$

b) Alt kesim frekansı = ?

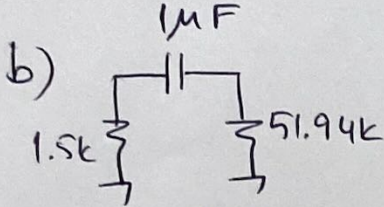
c) Üst kesim frekansı = ?

a) $A_v = -g_m \cdot (R_D \parallel R_L)$

$$A_v = -1.95\text{mS} (3.9\text{k} \parallel 5.6\text{k})$$

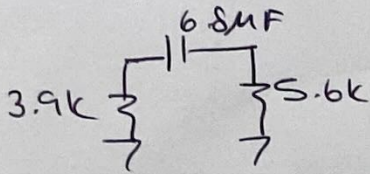
$$A_v = -4.485$$

5 puan



$$f_{a1} = \frac{1}{2\pi \cdot (1 \cdot 10^{-6}) (1.5\text{k} + 51.94\text{k})} = 2.9\text{Hz}$$

3P

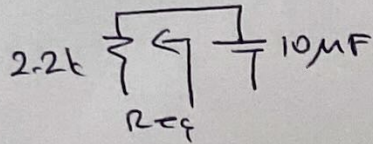


$$f_{a2} = \frac{1}{2\pi (6.8 \cdot 10^{-6}) (3.9\text{k} + 5.6\text{k})} = 2.46\text{Hz}$$

3P

$$f_a = 38.25\text{Hz}$$

1P



$$R_{cs} = 2.2\text{k} \parallel \frac{1}{1.95\text{mS}} \Rightarrow R_{cs} = 0.416\text{k}$$

$$f_{a3} = \frac{1}{2\pi (10 \cdot 10^{-6}) (0.416\text{k})} = 38.25\text{Hz}$$

3P

$$C_p = C_{w1} + C_i + C_{gs} =$$

$$C_i = C_{gd}(1 - A_v) = 43.84\text{pF}$$

$$C_i = 4\text{pF} + 43.84\text{pF} + 12\text{pF}$$

$$C_i = 59.84\text{pF}$$

$$R_i = 1.5\text{k} \parallel (51.94\text{k}) = 1.458\text{k}$$

$$f_{u1} = \frac{1}{2\pi (59.84 \cdot 10^{-12}) (1.458 \cdot 10^3)} = 1.82\text{MHz}$$

4P

$$C_o = C_{w2} + C_2 + C_{ds}$$

$$C_2 = C_{gd}(1 - \frac{1}{A_v}) = 9.79\text{pF}$$

$$C_o = 6\text{pF} + 9.79\text{pF} + 3\text{pF}$$

$$C_o = 18.79\text{pF}$$

$$R_o = 3.9\text{k} \parallel 5.6\text{k} = 2.3\text{k}$$

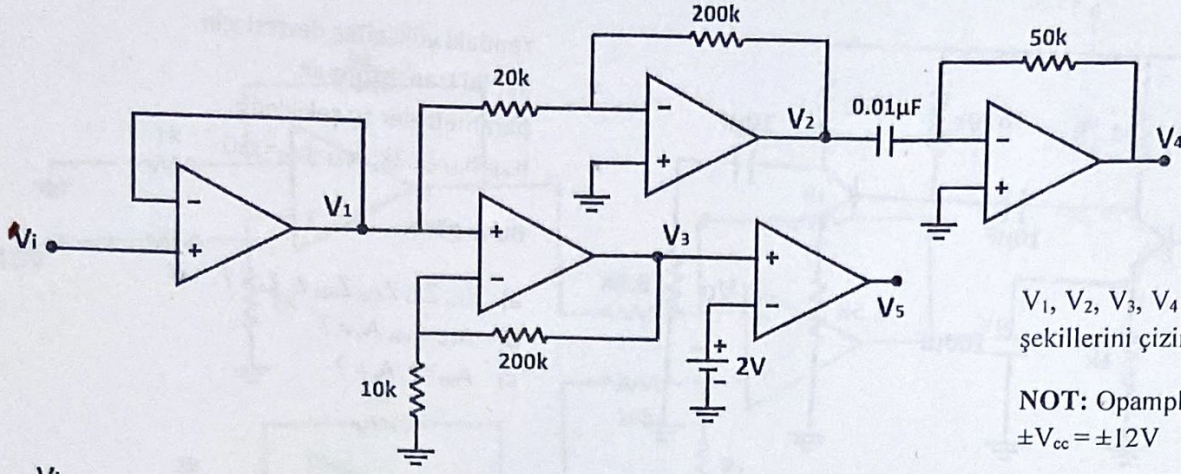
$$f_{u0} = \frac{1}{2\pi (18.79 \cdot 10^{-12}) (2.3 \cdot 10^3)} = 3.68\text{MHz}$$

4P

$$f_u = 1.82\text{MHz}$$

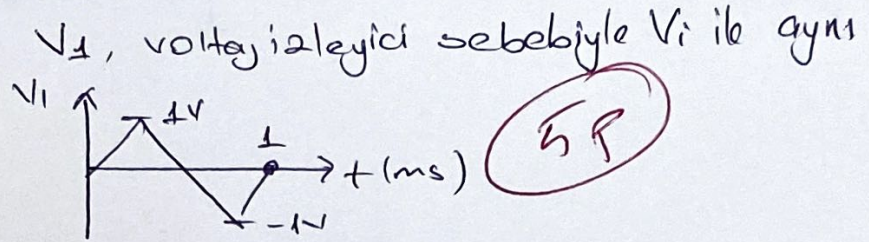
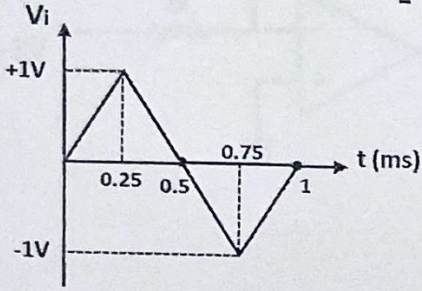
2P

S.4.



V_1, V_2, V_3, V_4 ve V_5 dalga şekillerini çiziniz.

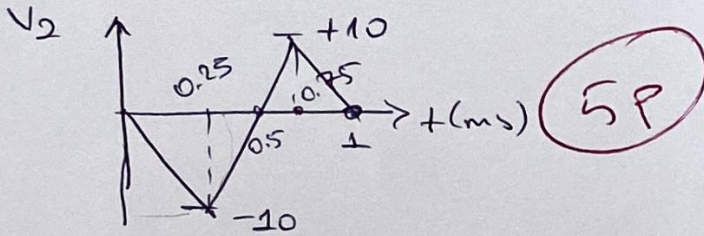
NOT: Opamplar İDEAL ve $\pm V_{cc} = \pm 12V$



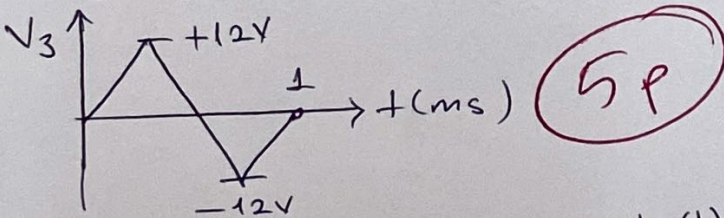
V_2 , faz çeviren (eviren) kuvvetlendirici.

$$V_2 = - \frac{200k}{20k} \cdot V_1$$

$$V_2 = -10 \cdot V_1$$



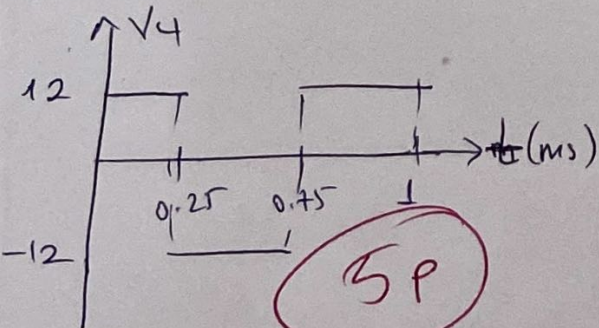
V_3 , faz çevirmeyen (evirmeyen) kuvvet. $V_3 = (1 + \frac{200k}{10k}) \cdot V_1$
($\pm V_{cc}$ 'yi geçeceği için kısıtlama var!) $\Leftarrow V_3 = 21 V_1$



V_4 , türev alma devresi

$$V_4(t) = -R \cdot C \cdot \frac{dV_2(t)}{dt} = -0.5 \cdot 10^{-3} \frac{dV_2(t)}{dt}$$

$0 < t < 0.25 \text{ ms}$ aralığında $V_2(t) = -40 \cdot 10^3 t \Rightarrow V_4(t) = 20$
 $0.25 \text{ ms} < t < 0.75 \text{ ms}$ " $V_2(t) = 40 \cdot 10^3 t - 10 \Rightarrow V_4(t) = -20$
 $0.75 \text{ ms} < t < 1 \text{ ms}$ " $V_2(t) = -40 \cdot 10^3 t + 40 \Rightarrow V_4(t) = 20$



V_5 , karşılaştırıcı (komp.)

$V_+ > V_- \Rightarrow +12V$ $V_+ < V_- \Rightarrow -12V$

